WORKED SOLUTION

STEP III 1990, Question 15

Probability and Statistics • Geometric Distributions

STEP III 1990, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q15

The key idea

Each throw shows A with probability $\frac{1}{4}$ and B with probability $\frac{3}{4}$, independently. So the event $\{M = m,\ N = n\}$ is one specific string of throws, and its probability is a product. That product splits into a factor depending only on m times a factor depending only on n, and this does almost all the work: the marginals are geometric, the variables are independent, and every later probability is a geometric series.

Solution

Write $p = \mathrm{P}(A) = \frac{3}{12} = \frac{1}{4}$ and $q = \mathrm{P}(B) = \frac{9}{12} = \frac{3}{4}$ for a single throw. Throws are independent.

The event $\{M = m,\ N = n\}$ happens exactly when the first $m + n + 1$ throws are

$$\underbrace{A\,A\cdots A}_{m}\ \underbrace{B\,B\cdots B}_{n}\ A,$$

so by independence, for $m \geq 0$ and $n \geq 1$,

$$\mathrm{P}(M = m,\ N = n) = \left(\tfrac{1}{4}\right)^m \left(\tfrac{3}{4}\right)^n \tfrac{1}{4} = \left(\tfrac{1}{4}\right)^{m+1} \left(\tfrac{3}{4}\right)^n.$$

(With probability 1 the sequence does eventually show a B and then an A, so these probabilities sum to 1; the marginals below confirm this.)

Part (i)

For the marginal of M, sum over $n \geq 1$, using $\sum_{n\geq 1} \left(\frac{3}{4}\right)^n = \dfrac{3/4}{1 - 3/4} = 3$:

$$\mathrm{P}(M = m) = \left(\tfrac{1}{4}\right)^{m+1} \cdot 3 = \tfrac{3}{4}\left(\tfrac{1}{4}\right)^m, \qquad m = 0, 1, 2, \ldots$$

For the marginal of N, sum over $m \geq 0$, using $\sum_{m\geq 0}\left(\frac{1}{4}\right)^{m+1} = \dfrac{1/4}{1-1/4} = \frac{1}{3}$:

$$\mathrm{P}(N=n) = \tfrac{1}{3}\left(\tfrac{3}{4}\right)^{n} = \tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1}, \qquad n = 1,2,3,\ldots$$

Both make sense directly: M counts As before the first B, and N is the waiting time for the first A once the B-run has started. Each sums to 1.

Part (ii)

We use $\sum_{k\geq 1} kx^{k-1} = \dfrac{1}{(1-x)^2}$ for $|x|<1$ (differentiate the geometric series). Then

$$\mathrm{E}(M) = \tfrac{3}{4}\sum_{m\geq 1} m\left(\tfrac{1}{4}\right)^{m} = \tfrac{3}{4}\cdot\tfrac{1}{4}\cdot\frac{1}{(3/4)^2} = \tfrac{1}{3},$$

$$\mathrm{E}(N) = \tfrac{1}{4}\sum_{n\geq 1} n\left(\tfrac{3}{4}\right)^{n-1} = \tfrac{1}{4}\cdot\frac{1}{(1/4)^2} = 4.$$

Independence

Multiplying the marginals,

$$\mathrm{P}(M=m)\,\mathrm{P}(N=n) = \tfrac{3}{4}\left(\tfrac{1}{4}\right)^{m}\cdot\tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1} = \left(\tfrac{1}{4}\right)^{m+1}\left(\tfrac{3}{4}\right)^{n} = \mathrm{P}(M=m,\ N=n)$$

for every $m \geq 0$, $n \geq 1$. So M and N *are* independent.

Ask yourself Surely the length of the A-run affects the B-run? Why doesn't it?

The A-run ends at the first B, and from that throw on the process starts afresh: the remaining throws are independent of everything before. The length of the B-run depends only on those later throws, so knowing M tells us nothing about N. The factorisation of the joint probability is the algebraic form of this.

The probabilities involving N and M

For an integer $k \geq 1$, sum the tail of the geometric distribution:

$$\mathrm{P}(N>k) = \sum_{n\geq k+1}\tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1} = \tfrac{1}{4}\cdot\frac{(3/4)^k}{1-3/4} = \left(\tfrac{3}{4}\right)^{k}.$$

This formula also gives $\mathrm{P}(N>0)=1$, which is correct since $N \geq 1$.

For $\mathrm{P}(N>M)$, condition on M and use independence:

$$\begin{aligned}\mathrm{P}(N>M) &= \sum_{m\geq 0}\mathrm{P}(M=m)\,\mathrm{P}(N>m) = \sum_{m\geq 0}\tfrac{3}{4}\left(\tfrac{1}{4}\right)^{m}\left(\tfrac{3}{4}\right)^{m}\\ &= \tfrac{3}{4}\sum_{m\geq 0}\left(\tfrac{3}{16}\right)^{m} = \tfrac{3}{4}\cdot\tfrac{16}{13} = \tfrac{12}{13}.\end{aligned}$$

For $\mathrm{P}(N = M)$ the common value must be at least 1, since $N \geq 1$:

$$\mathrm{P}(N = M) = \sum_{m \geq 1} \tfrac{3}{4} \left(\tfrac{1}{4}\right)^m \cdot \tfrac{1}{4} \left(\tfrac{3}{4}\right)^{m-1} = \tfrac{3}{64} \sum_{m \geq 1} \left(\tfrac{3}{16}\right)^{m-1}$$
$$= \tfrac{3}{64} \cdot \tfrac{16}{13} = \tfrac{3}{52}.$$

Finally, the three events $N > M$, $N = M$, $N < M$ are exclusive and exhaustive, so

$$\mathrm{P}(N < M) = 1 - \tfrac{12}{13} - \tfrac{3}{52} = \tfrac{52-48-3}{52} = \tfrac{1}{52},$$

as required.

Checks

- The means fit intuition: A is rare, so the opening A-run is short ($\tfrac{1}{3}$ on average), while we wait on average $1/p = 4$ throws for an A to end the B-run.
- Directly, $\mathrm{P}(M > n) = \left(\tfrac{1}{4}\right)^{n+1}$, so $\mathrm{P}(N < M) = \sum_{n \geq 1} \tfrac{1}{4} \left(\tfrac{3}{4}\right)^{n-1} \left(\tfrac{1}{4}\right)^{n+1} = \tfrac{1}{64} \sum_{n \geq 1} \left(\tfrac{3}{16}\right)^{n-1} = \tfrac{1}{64} \cdot \tfrac{16}{13} = \tfrac{1}{52}$, agreeing with the subtraction. The direct reason for $\mathrm{P}(N > k) = \left(\tfrac{3}{4}\right)^k$: after the first B, the next k throws must all be B.
- Exact summation of the joint distribution gives $\mathrm{P}(N > M) = 0.92308$, $\mathrm{P}(N = M) = 0.05769$, $\mathrm{P}(N < M) = 0.01923$; a simulation of 200 000 series of throws gave 0.923, 0.058, 0.019 and sample means 0.334 and 3.99.

Common slips

- Including $m = 0$ in the sum for $\mathrm{P}(N = M)$: N can never be 0.
- Forgetting that M can be 0 (the first throw may be B), which shifts the marginal of M and gives the wrong mean.

Beyond STEP

The run lengths here are independent because a sequence of independent trials "restarts" at any time determined only by the past, such as the first B. This is the *strong Markov property*, and it underlies renewal theory and the analysis of Markov chains, which are first- and second-year undergraduate topics. Grimmett and Welsh, *Probability: An Introduction*, treats geometric distributions, independence and Markov chains at the right level.

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